

Autonomous Vacuum Cleaner

Report: EE396 Design Lab

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1 Introduction

This report details the design and development of an Autonomous Vacuum Cleaner, a prototype created to address this need as part of the EE396 Design Lab.

The primary objective of this project is to conceptualize and build a self-navigating robotic system capable of autonomous floor cleaning without human intervention. Rather than relying on manual operation, the proposed device uses real-time environmental feedback to make navigational decisions while actively sweeping and capturing debris.

The remainder of this report outlines the necessity and specific problem statement, details the core design principles and hardware trade-offs, and discusses the physical implementation. Finally, it presents the testing results and explores potential future prospects for refining the prototype.

2 Design Principles and Trade-offs

The mechanical and electrical design of the autonomous vacuum cleaner required several iterative adjustments to balance performance, power constraints, and structural stability. The core design principles and the resulting trade-offs are detailed below:

1. Chassis Design and Locomotion Strategy A circular acrylic chassis was chosen over a rectangular design to prevent the robot from getting stuck in corners, thereby enhancing its rotational maneuverability. Moreover, the 12V supply battery was hand held to prevent weight load on N20 motors. For locomotion, we implemented a differential drive system utilizing two rear robotic wheels, stabilized by a single front caster wheel.

- **Trade-off:** Opting for a single caster wheel in the front instead of a four-wheel drive system significantly reduced the overall weight and power consumption while allowing for a zero-degree turning radius.

2. Drivetrain Assembly and Mechanical Stabilization Initial testing revealed significant mechanical flaws in the drivetrain: the rear wheels exhibited severe wobble during rotation, and the DC motors outputted asymmetrical speeds.

- **Implementation:** To counteract the wobble and stabilize the rotational axis, we custom 3D-printed coupling shafts to securely connect the wheels to the DC motors. Furthermore, M-Seal epoxy putty was applied to the mounts to rigidly lock the assembly in place, effectively eliminating the vibrations. Also to prevent skidding, we had put two rubber bands over the wheel circumference.

3. Power Distribution and Motor Control Early prototypes experienced motor stalling when the rear wheels made contact with the ground under load. Initially, the L293D motor driver was relying heavily on logic-level power, which caused severe voltage drops and hampered performance.

- **Implementation:** To resolve this, we bypassed the Arduino's onboard 5V supply for driving heavy loads. Instead, the 12V main supply was routed through a dedicated buck converter to deliver a stable, high-current operating voltage directly to the L293D motor driver.

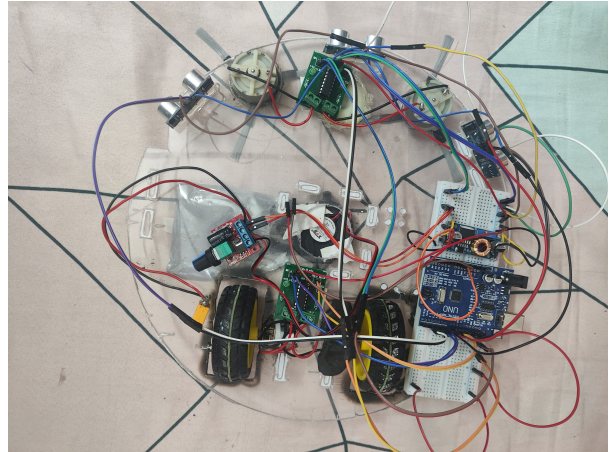
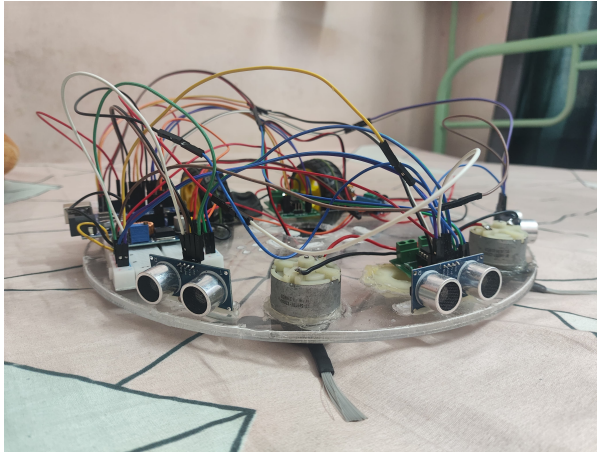
- **Trade-off:** While this added slight complexity to the wiring harness, decoupling the motor power from the microcontroller’s logic supply ensured consistent torque and prevented unexpected system resets.

4. Suction Mechanism and Placement A critical limitation encountered was the underpowered nature of the primary suction motor. Upgrading to a high-wattage vacuum motor was not feasible, as it would have drastically exceeded the current limits of our 12V power architecture and severely drained the battery.

- **Implementation:** To maximize the efficiency of the existing low-power motor, the vacuum suction unit was mounted at the middle-bottom of the lowest acrylic sheet. By minimizing the clearance between the suction intake and the floor, we maximized the static pressure.
- **Trade-off:** As a compromise for retaining the low-power electrical architecture, the device’s cleaning capability is currently limited to lightweight debris; it is highly effective at pulling fine particulate matter, such as standard dust and chalk powder, but cannot handle heavy or dense debris.

5. Sweeping Mechanism and Brush Friction During initial ground testing, the original motors driving the front rotary brushes lacked the necessary torque. The friction generated by the brushes making full contact with the floor caused the motors to frequently jam and stall, impeding the robot’s overall forward momentum.

- **Implementation:** We replaced the stock brush motors with heavy-duty, high-torque DC motors to ensure consistent rotation even under physical resistance. Additionally, we manually trimmed the bristle length of both brushes so they glide just above the floor rather than pressing firmly into it.
- **Trade-off:** Cutting the brushes slightly reduced their direct scrubbing pressure against the floor. However, this was a necessary compromise to prevent motor stalling and battery drain; the brushes now effectively guide loose debris toward the central suction intake without creating excess drag that would hinder the robot’s locomotion.



3 Implementation

The physical assembly of the autonomous vacuum cleaner focused on cleanly integrating the mechanical, electromechanical, and sensor systems into a single functional prototype. The step-by-step implementation of the device is detailed below:

- **Chassis and Structural Assembly:** The foundation of the robot was constructed using an acrylic sheet. We deliberately maintained an open-chassis architecture—omitting a top cover—to accommodate several peripheral components and structural elements that extended slightly beyond the circular circumference. This open design also facilitated easier access for real-time hardware debugging.

- **Sensors and Locomotion:** To enable spatial awareness, an array of three ultrasonic sensors was securely mounted along the front perimeter, providing a wide detection angle for incoming obstacles. For mobility, the custom-stabilized rear drive wheels and the single front caster wheel were mounted to the underside of the acrylic base, ensuring balanced weight distribution.
- **Cleaning Module Integration:** The sweeping mechanism, consisting of the two front rotary brushes, was mounted at the leading edge of the chassis. Behind the brushes, the primary vacuum module was installed. This consisted of a central circular suction pipe positioned close to the floor to maximize static pull, which routes collected debris directly into an attached dust bag.
- **Electronics and Power Distribution:** The system’s logic and spatial processing are handled by a central Arduino controller. To manage the physical movement, an L293D motor driver was wired to simultaneously control the two rear drive wheels and the two front sweeping brushes. The central vacuum operates via its own dedicated suction pipe controller board. The entire electromechanical setup is powered by a central 12V battery supply.
- **Arduino Code:** The code part can be found here: [Project Repository](#)

4 Results and Testing

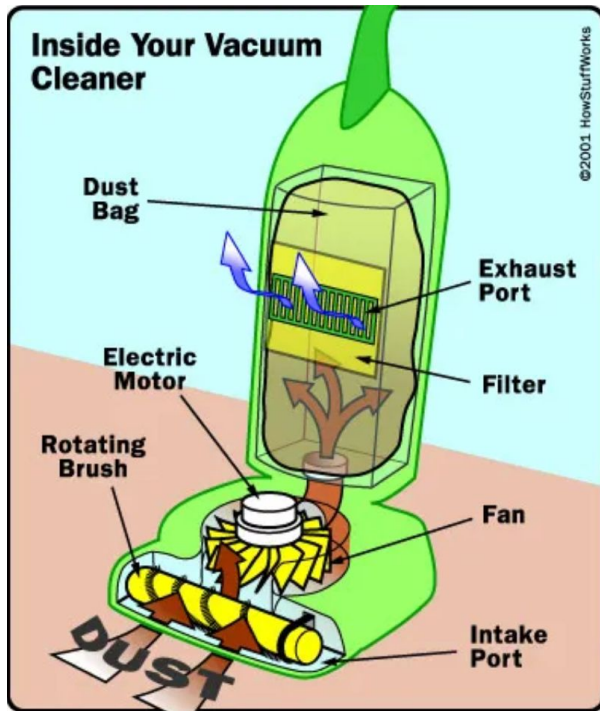
The autonomous vacuum cleaner prototype was subjected to operational floor testing to evaluate its navigation capabilities and cleaning efficacy.

- **Cleaning Performance:** As anticipated during the design phase, the system successfully managed lightweight particulate matter. Testing specifically with standard floor dust and chalk dust pieces confirmed that the low-clearance central suction pipe effectively pulled the debris into the collection bag. Furthermore, the modified, trimmed front brushes successfully swept the chalk dust toward the central intake without causing the heavy-duty motors to stall.
- **Navigation and Mobility:** The array of three front-facing ultrasonic sensors successfully detected physical impediments, allowing the Arduino logic to halt the drive motors and prevent collisions. The combination of the stabilized rear drive wheels and the front caster wheel provided smooth, unimpeded maneuverability during the test runs.

5 Future Prospects

While the current prototype effectively validates the core electromechanical concepts, several improvements could be implemented in future iterations to enhance performance and autonomy:

- **Power and Suction Upgrades:** A primary focus would be upgrading the 12V power architecture to support a higher-wattage vacuum motor. This would allow the robot to transition from pulling fine dust to handling heavier or denser debris.
- **Advanced Navigation:** Replacing the simple reactive obstacle avoidance with room mapping capabilities or floor-type detection would significantly optimize cleaning routes and efficiency.
- **Enclosure Design:** Developing a custom, lightweight 3D printed top cover would enclose the currently exposed open-chassis circuitry, protecting the Arduino, motor drivers, and buck converter from the very dust the robot is collecting.
- **True Autonomy:** Integrating an automatic self-charging dock would allow the system to recharge its battery without human intervention, enabling continuous operation.



6 Conclusion

The development of this autonomous vacuum cleaner prototype successfully demonstrates the integration of sensor-based navigation and electromechanical cleaning systems. Despite initial challenges—such as drive wheel wobble, logic-level power drops, and brush motor stalling—strategic engineering compromises and physical modifications resulted in a better functional device. By implementing custom 3D-printed stabilization shafts, a dedicated buck converter for power decoupling, and optimized brush lengths, the system effectively navigates its environment and reliably cleans lightweight debris.

Acknowledgments

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